

EMPLOYMENT

Massachusetts Institute of Technology
 C. L. E. Moore instructor, Mentor: Wei Zhang

Cambridge, MA
 Fall 2026 - Spring 2029

EDUCATION

University of Wisconsin - Madison

Ph.D. in Mathematics, Advisors: Michael Rapoport (primary), Tonghai Yang (secondary)

Madison, WI

May 2026

University of Wisconsin - Madison

M.A. in Mathematics

Madison, WI

May 2021

Zhejiang University

B.S. in Mathematics

Hangzhou, ZJ

May 2020

PREPRINTS

1. Yu Luo, Michael Rapoport and Wei Zhang, “More regular formal moduli spaces and arithmetic transfer conjectures: the ramified quadratic case”, [submitted](#)
2. Yu Luo, “Kudla-Rapoport conjecture for unramified maximal parahoric level”, [submitted](#)
3. Yu Luo, Andreas Mihatsch and Zhiyu Zhang, “Unitary Shimura varieties at ramified primes and arithmetic transfer”, [submitted](#)
4. Qiao He, Yu Luo and Yousheng Shi, “The basic locus of ramified unitary Rapoport-Zink spaces at maximal vertex level”, [submitted](#)
5. Qiao He, Yu Luo and Yousheng Shi, “Regular models of ramified unitary Shimura varieties at maximal parahoric level”, [submitted](#)

PUBLICATIONS

1. Yu Luo, “On the moduli description of ramified unitary local models of signature $(n-1, 1)$ ”, **Math. Ann.** (2025)

INVITED TALKS

- Arithmetic Siegel-Weil formula at maximal parahoric level
Caltech Number Theory Seminar, Pasadena, California, USA April 2026
- Geometric and arithmetic Siegel-Weil formula
Johns Hopkins Junior Number Theory Days, Baltimore, Maryland, USA Feb 2026
- A new proof of the arithmetic Siegel-Weil formula
Joint Mathematics Meetings 2026, Washington, D.C., USA Jan 2026
- On unitary Rapoport-Zink space
Universität Münster, Münster, Germany Nov 2025

- Proof of the arithmetic Siegel-Weil formula Nov 2025
University of Duisburg-Essen, Essen, Germany
- A new proof of the arithmetic Siegel-Weil formula Nov 2025
Universität Münster, Münster, Germany
- A new proof of the arithmetic Siegel-Weil formula Sept 2025
POSTECH, Pohang, South Korea
- A new proof of the arithmetic Siegel-Weil formula June 2025
Symposium on Number Theory and Representation Theory, IX, Hangzhou, China
- A new proof of the arithmetic Siegel-Weil formula May 2025
Zhejiang University
- A new proof of the arithmetic Siegel-Weil formula Feb 2025
University of Wisconsin - Madison
- Comparison of absolute and relative unitary Rapoport-Zink spaces Oct 2024
Massachusetts Institute of Technology (Learning seminar on Arithmetic Inner Product Formula)
- On the Moduli Description of a Class of Unitary Shimura Varieties Dec 2023
University of Wisconsin - Madison
- On the Moduli Description of a Class of Unitary Shimura Varieties May 2023
Zhejiang University
- On the Moduli Description of a Class of Unitary Shimura Varieties May 2023
Michigan State University

SERVICES

- Co-organizer of the Langlands-Kottwitz method reading Seminar 2025 Fall - 2026 Spring
University of Wisconsin - Madison
- Co-organizer of the Relative trace formula reading Seminar 2024 Fall - 2025 Spring
University of Wisconsin - Madison
- Organizer of the Gross-Zagier formula reading Seminar 2024 Spring
University of Wisconsin - Madison
- Co-organizer of GAGS - Graduate Algebraic Geometry Seminar 2022 Spring-2025 Spring
University of Wisconsin - Madison
- Mentor at Undergraduate Directed Reading Program 2022 Spring
Mentor of Baoyi WANG (Currently in Columbia University) University of Wisconsin - Madison
- Mentor of Master Program in Math Department 2021 Fall
Mentor of Ruoxi LI (Currently in UIUC Math PhD), Weiman YUAN (Currently in Vanderbilt University Bio, Engineering), Aditya Phukon.
- Organizer of Algebraic Geometry Learning Seminar 2020 Spring
University of Wisconsin - Madison
- Organizer of Commutative Algebra Learning Seminar 2019 Fall
University of Wisconsin - Madison
- Organizer of Undergraduate Algebraic Geometry Seminar 2019 Spring
Zhejiang University

TEACHING

- Math 340 - Linear algebra - TA 2026 Spring
University of Wisconsin - Madison
- Math 221 - Calculus 1 - TA 2025 Fall
University of Wisconsin - Madison
- Math 475 - Introduction to Combinatorics - TA 2025 Summer
University of Wisconsin - Madison
- Math 435 - Introduction to Cryptography - TA 2025 Summer
University of Wisconsin - Madison
- Math 340 - Linear Algebra - TA 2025 Spring
University of Wisconsin - Madison
- Math 112 - Algebra - Lecturer 2024 Fall
University of Wisconsin - Madison
- Math 320 - Linear Algebra and ODE - TA 2024 Spring
University of Wisconsin - Madison
- Math 340 - Linear Algebra - TA 2023 Fall
University of Wisconsin - Madison
- Math 340 - Linear Algebra - TA 2023 Spring
University of Wisconsin - Madison
- Math 234 - Calculus 3 - TA 2022 Fall
University of Wisconsin - Madison
- Math 240 - Discrete Math - TA 2024 Summer
University of Wisconsin - Madison
- Math 222 - Calculus 2 - TA 2022 Spring
University of Wisconsin - Madison
- Math 221 - Calculus 1 - TA 2021 Fall
University of Wisconsin - Madison

HONORS AND AWARDS

- ICM 2026 Travel Grant Program 2026 Summer
- Excellence in Mathematical Research Award - University of Wisconsin, Madison 2025 Fall
- Government Scholarship - Zhejiang Government 2017–2018
- Government Scholarship - Zhejiang Government 2016–2017